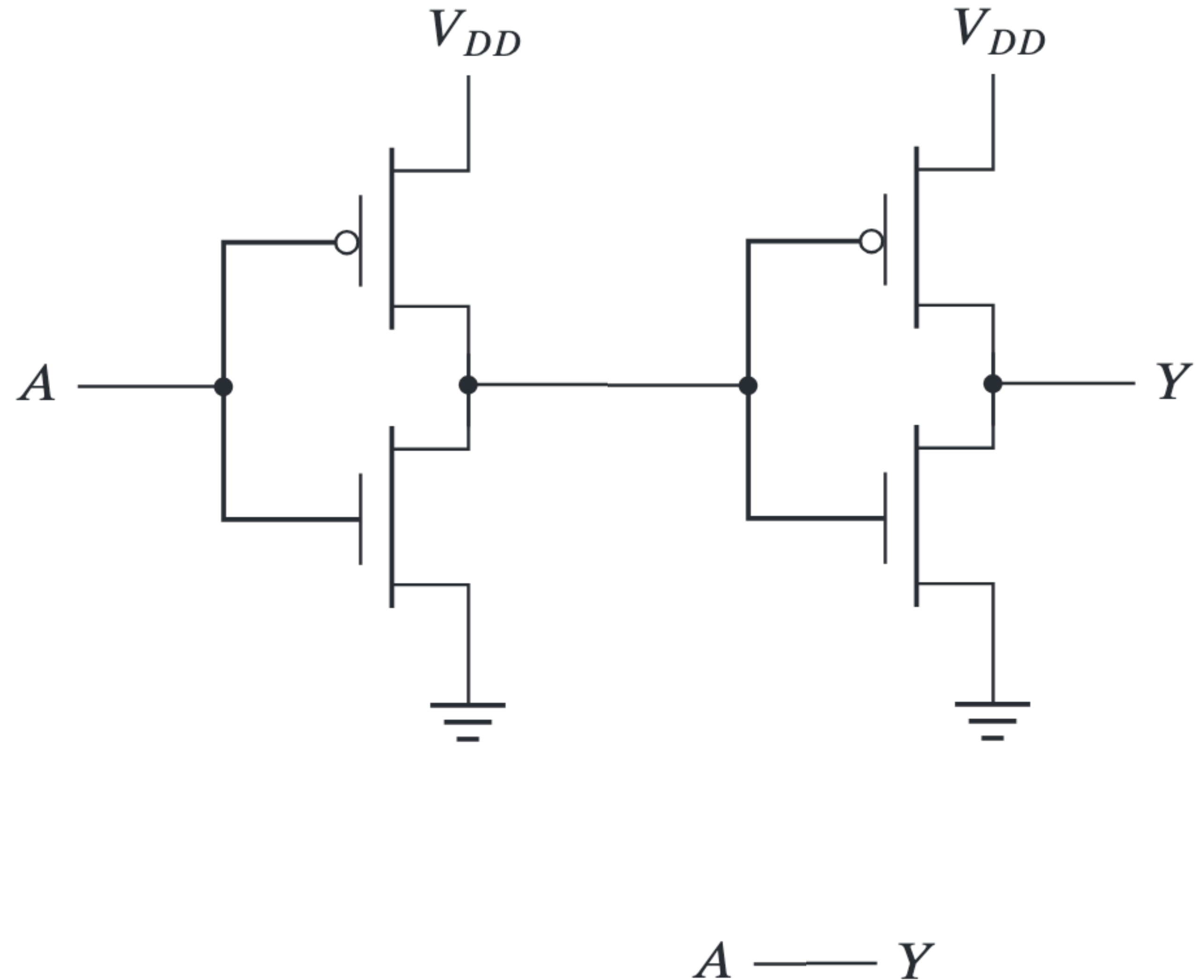


Boolean Algebra and Logic Gates

Abhilash Jindal

Introduction

- Finding simpler and cheaper, but equivalent, realisations of a circuit can reduce overall cost
- Boolean algebra to optimize simple circuits
- $Y = (A')' = A$



Binary Operators

Closure. $\forall a, b \in S : \exists c \in S : a \square b = c$

- Commutative: $\forall x, y \in S : x \square y = y \square x$
- Associative: $\forall x, y \in S : (x \square y) \square z = x \square (y \square z)$
- Identity element: $\exists e \in S : \forall x \in S : e \square x = x \square e = x$
- Inverses: $\forall x \in S : \exists y \in S : x \square y = e$
- Distributive: $\forall x, y, z \in S : x \square (y \diamond z) = (x \square y) \diamond (x \square z)$

$(B, +, *)$ is a Boolean Algebra

Huntington Postulates

- Set of elements B and two binary operators $+$ and $*$ (or $\cdot$)
 - B is closed on $+$ and $*$
 - 0 is identity with respect to $+$ and 1 is identity with respect to $*$
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
 - Both $+$ and $*$ are commutative
 - $*$ distributes over $+$ and $+$ distributes over $*$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $x + (y \cdot z) = (x + y) \cdot (x + z)$
 - $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$

Is $(\mathbb{R}, +, *)$ a Boolean Algebra?

Huntington Postulates

- Set of elements B and two binary operators $+$ and $*$ (or $\cdot$)
 - B is closed on $+$ and $*$
 - 0 is identity with respect to $+$ and 1 is identity with respect to $*$
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
 - Both $+$ and $*$ are commutative
 - $*$ distributes over $+$ and $+$ distributes over $*$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $x + (y \cdot z) = (x + y) \cdot (x + z)$
 - $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$
- + does not distributes over * !
- + and * do not have inverses.
- and / are not defined!
- Complements do not exist

Is $(\mathcal{P}(\{a, b\}), \cup, \cap)$ a Boolean Algebra?

Huntington Postulates

0 is $\{\}$. 1 is $\{a, b\}$

- B is closed on $+$ and $*$
- 0 is identity with respect to $+$ and 1 is identity with respect to $*$
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
- Both $+$ and $*$ are commutative
- $*$ distributes over $+$ and $+$ distributes over $*$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $x + (y \cdot z) = (x + y) \cdot (x + z)$
- $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$

x' is $\{a, b\} \setminus x$

Theorems and properties

- $x + x = x$

- $x + x =$

- $(x + x) * 1$ $y * 1 = y$

Identity

- $(x + x) * (x + x')$ $(x + x') = 1$

Complement

- $x + x * x'$ $(x+y)*(x+z) = x + (y * z)$

+ Distributes over *

- $x + 0$ $x * x' = 0$

Complement

- x $y + 0 = y$

Identity

Theorems and properties

- $x + 1 = 1$

- $x + 1 =$

- $1 * (x + 1)$

$$1 * y = y$$

Identity

- $(x + x') * (x + 1)$

$$(x + x') = 1$$

Complement

- $x + x'^*1$

$$(x+y)*(x+z) = x + (y * z)$$

+ Distributes over *

- $x + x'$

$$x'^*1 = x'$$

Identity

- 1

$$x + x' = 1$$

Complement

Duality of Boolean Algebra

Huntington Postulates

- Set of elements B and two binary operators $+$ and $*$ (or $\cdot$)
 - B is closed on $+$ and $*$
 - 0 is identity with respect to $+$ and 1 is identity with respect to $*$
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
 - Both $+$ and $*$ are commutative
 - $*$ distributes over $+$ and $+$ distributes over $*$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $x + (y \cdot z) = (x + y) \cdot (x + z)$
 - $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$

Swap $+$ and $*$. Swap 0 and 1

Is $(\mathcal{P}(\{a, b\}), \cap, \cup)$ a Boolean Algebra?

Huntington Postulates

0 is $\{a, b\}$. 1 is $\{\}$

- B is closed on $+$ and $*$
- 0 is identity with respect to $+$ and 1 is identity with respect to $*$
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
- Both $+$ and $*$ are commutative
- $*$ distributes over $+$ and $+$ distributes over $*$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $x + (y \cdot z) = (x + y) \cdot (x + z)$
- $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$

x' is $\{a, b\} \setminus x$

Dual Theorems

Swap + and *. Swap 0 and 1

(a) $x + 0 = x$

(b) $x \cdot 1 = x$

(a) $x + x' = 1$

(b) $x \cdot x' = 0$

(a) $x + x = x$

(b) $x \cdot x = x$

(a) $x + 1 = 1$

(b) $x \cdot 0 = 0$

$(x')' = x$

(a) $x + y = y + x$

(b) $xy = yx$

(a) $x + (y + z) = (x + y) + z$

(b) $x(yz) = (xy)z$

(a) $x(y + z) = xy + xz$

(b) $x + yz = (x + y)(x + z)$

(a) $(x + y)' = x'y'$

(b) $(xy)' = x' + y'$

(a) $x + xy = x$

(b) $x(x + y) = x$

Dual theorem proofs

Swap + and *. Swap 0 and 1

- $x * x = x$

- $x * x =$

- $(x * x) + 0$ $y + 0 = y$

- $(x * x) + (x * x')$ $(x * x') = 0$

- $x * (x + x')$ $(x * y) + (x * z) = x * (y + z)$

- $x * 1$ $x + x' = 1$

- x $y * 1 = y$

Identity

Complement

* Distributes over +

Complement

Identity

Absorption theorem

- $x + x^*y = x$
- $x + x^*y =$
 - $x^*1 + x^*y$
 - $x^*(1 + y)$
 - x^*1
 - x

- $x^*(x + y) = x$
- $x^*(x + y) =$
 - $(x + 0)^*(x + y)$
 - $x + (0^*y)$
 - $x + 0$
 - x

Is $(\{0,1\}, +, \cdot)$ a Boolean Algebra?

Huntington Postulates

- **B is closed on + and ***
- 0 is identity with respect to + and 1 is identity with respect to *
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
- Both + and * are commutative
- * distributes over + and + distributes over *
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and
 $x + (y \cdot z) = (x + y) \cdot (x + z)$
- $\forall x \in B$, there exists a complement $x' \in B$ such that
 $x + x' = 1$ and $x * x' = 0$

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

Is $(\{0,1\}, +, \cdot)$ a Boolean Algebra?

Huntington Postulates

- B is closed on $+$ and $*$
- **0 is identity with respect to $+$ and 1 is identity with respect to $*$**
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
- Both $+$ and $*$ are commutative
- $*$ distributes over $+$ and $+$ distributes over $*$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and
 $x + (y \cdot z) = (x + y) \cdot (x + z)$
- $\forall x \in B$, there exists a complement $x' \in B$ such that
 $x + x' = 1$ and $x * x' = 0$

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

Is $(\{0,1\}, +, \cdot)$ a Boolean Algebra?

Huntington Postulates

- B is closed on $+$ and $\cdot$
- 0 is identity with respect to $+$ and 1 is identity with respect to $\cdot$
 $0 + x = x + 0 = x$ and $x \cdot 1 = 1 \cdot x = x$
- **Both $+$ and $\cdot$ are commutative**
- $\cdot$ distributes over $+$ and $+$ distributes over $\cdot$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and
 $x + (y \cdot z) = (x + y) \cdot (x + z)$
- $\forall x \in B$, there exists a complement $x' \in B$ such that
 $x + x' = 1$ and $x \cdot x' = 0$

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

Is $(\{0,1\}, +, \cdot)$ a Boolean Algebra?

Huntington Postulates

- B is closed on $+$ and $\cdot$
- 0 is identity with respect to $+$ and 1 is identity with respect to $\cdot$
 $0 + x = x + 0 = x$ and $x \cdot 1 = 1 \cdot x = x$
- Both $+$ and $\cdot$ are commutative
- **$\cdot$ distributes over $+$ and $+$ distributes over $\cdot$**
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and
 $x + (y \cdot z) = (x + y) \cdot (x + z)$
- $\forall x \in B$, there exists a complement $x' \in B$ such that
 $x + x' = 1$ and $x \cdot x' = 0$

x	y	z	$y + z$	$x \cdot (y + z)$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	1	1
1	0	1
1	1	1

Is $(\{0,1\}, +, \cdot)$ a Boolean Algebra?

Huntington Postulates

- B is closed on $+$ and $\cdot$
- 0 is identity with respect to $+$ and 1 is identity with respect to $\cdot$
 $0 + x = x + 0 = x$ and $x \cdot 1 = 1 \cdot x = x$
- Both $+$ and $\cdot$ are commutative
- $\cdot$ distributes over $+$ and $+$ distributes over $\cdot$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and
 $x + (y \cdot z) = (x + y) \cdot (x + z)$
- $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x \cdot x' = 0$

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

x	x'
0	1
1	0

Proving by truth tables

Absorption Theorem

- $x + x^*y = x$
- Do both sides yield identical results for all possible combinations of the variables involved?

<i>x</i>	<i>y</i>
0	0
0	1
1	0
1	1

<i>xy</i>	<i>x + xy</i>
0	0
0	0
0	1
1	1

Proving by truth tables

DeMorgan's Law

- $(x+y)' = x'y'$
- Do both sides yield identical results for all possible combinations of the variables involved?

x	y	$x + y$	$(x + y)'$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

x'	y'	$x'y'$
1	1	1
1	0	0
0	1	0
0	0	0

Cancellation law?

- If $x+z = y+z$ then $x=y$
- Holds in real numbers, $+$, $*$. But not in Boolean Algebra as $0+1=1+1$, but $0 \neq 1$

Big Idea

Switching algebra

- “One of the most important masters thesis ever written.. a landmark in that it helped to change digital circuit design from an art to science”

A Symbolic Analysis of Relay and Switching Circuits*

*Claude E. Shannon***

I. Introduction

In the control and protective circuits of complex electrical systems it is frequently necessary to make intricate interconnections of relay contacts and switches. Examples of these circuits occur in automatic telephone exchanges, industrial motor-control equipment, and in almost any circuits designed to perform complex operations automatically. In this paper a mathematical analysis of certain of the properties of such networks will be made. Particular attention will be given to the problem of network synthesis. Given certain characteristics, it is required to find a circuit incorporating these characteristics. The solution of this type of problem is not unique and methods of finding those particular circuits requiring the least number of relay contacts and switch blades will be studied. Methods will also be described for finding any number of circuits equivalent to a given circuit in all operating characteristics. It will be shown that several of the well-known theorems on impedance networks have roughly analogous theorems in relay circuits. Notable among these are the delta-wye and star-mesh transformations, and the duality theorem.

The method of attack on these problems may be described briefly as follows: any circuit is represented by a set of equations, the terms of the equations corresponding to the various relays and switches in the circuit. A calculus is developed for manipulating these equations by simple mathematical processes, most of which are similar to ordinary algebraic algorithms. This calculus is shown to be exactly analogous to the calculus of propositions used in the symbolic study of logic. For the synthesis problem the desired characteristics are first written as a system of equations, and the equations are then manipulated into the form representing the simplest circuit. The circuit may then be immediately drawn from the equations. By this method it is always possible to find the simplest circuit containing only series and parallel connections, and in some cases the simplest circuit containing any type of connection.

Point of view is worth 80 IQ points — Alan Kay

What must be the square which, when increased by ten of its own roots, amounts to thirty-nine?

Halve the number of roots, which in the present instance yields five. Multiply this by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of roots, which is five; the remainder is three. This is the root of the square which you sought, and the square itself is nine. — Al Khwarizmi (830 CE)

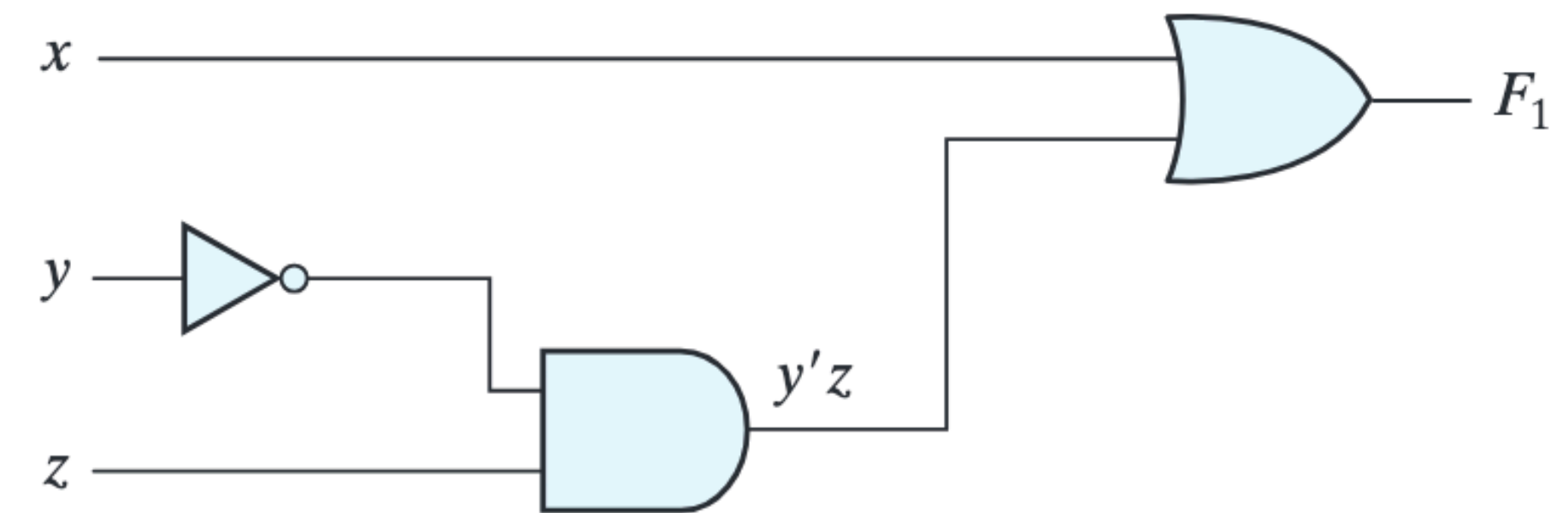
$$* x^2 + 10x = 39$$

$$* x = \frac{-10 \pm \sqrt{10^2 + 4 * 39}}{2}$$

$$* x = \frac{-10 + 16}{2}$$

Schematic

- Circuit connects input wires to output wires
- Can think of it as a boolean function: variables of the function are inputs and the function result is the output
- $F_1 = x + y'z$



Boolean Function

- Algebraic expression consisting of binary variables, constants 0 and 1, and logic operation symbols.
- For a given value of input, the output is either 0 or 1.
 - $F_1 = x + y'z$.
1 when $x=1$ or when $yz=01$

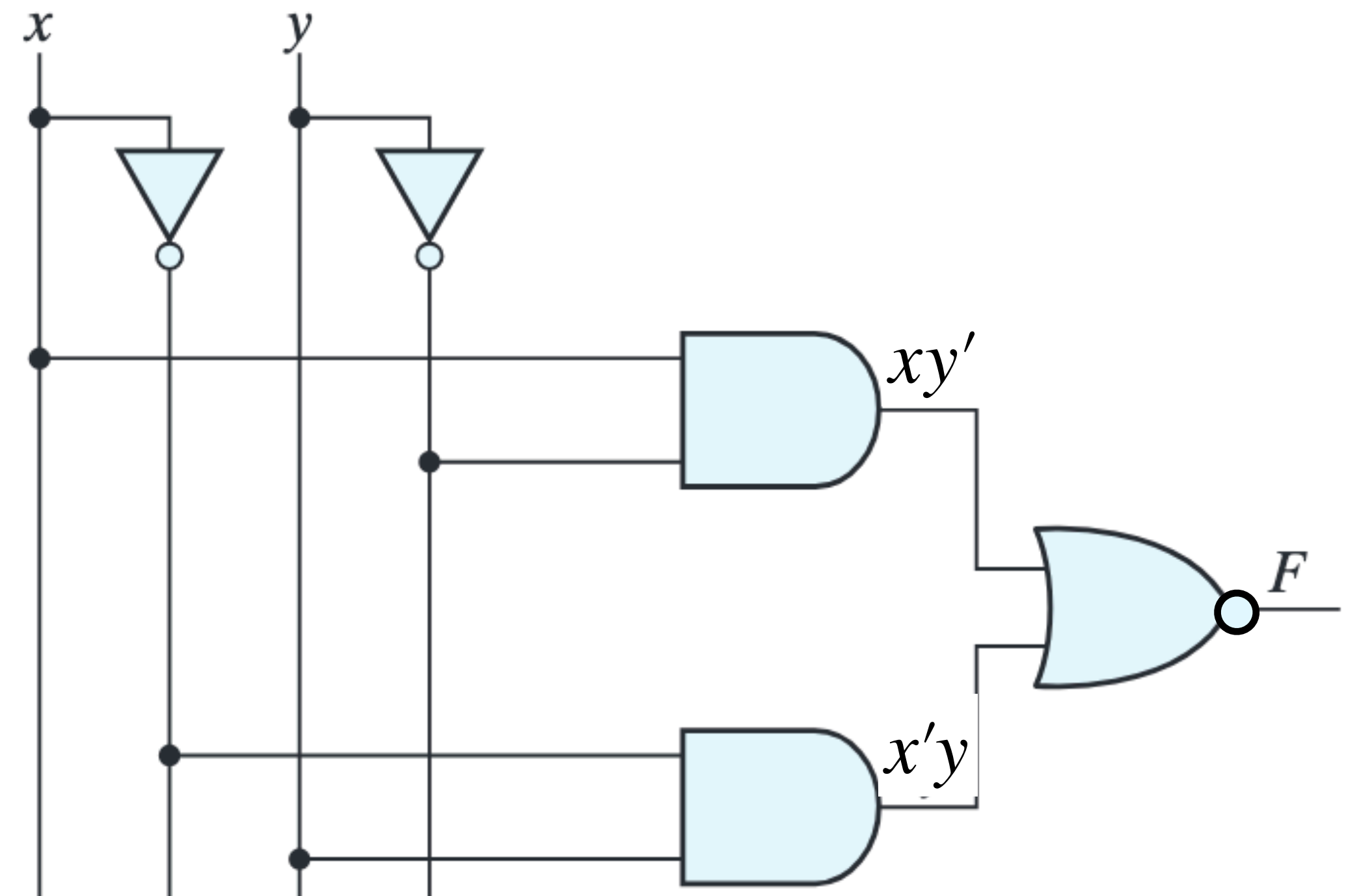
x	y	z	F_1
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Complement of a function

- Complement of a function F' : interchange 1 to 0 and 0 to 1 in the output of F .
- DeMorgan's law:
 - $(x + y + z)' = x'y'z'$
 - $(xyz)' = x' + y' + z'$
- $F_1 = x'yz' + x'y'z$
 - $F'_1 = (x + y' + z)(x + y + z')$

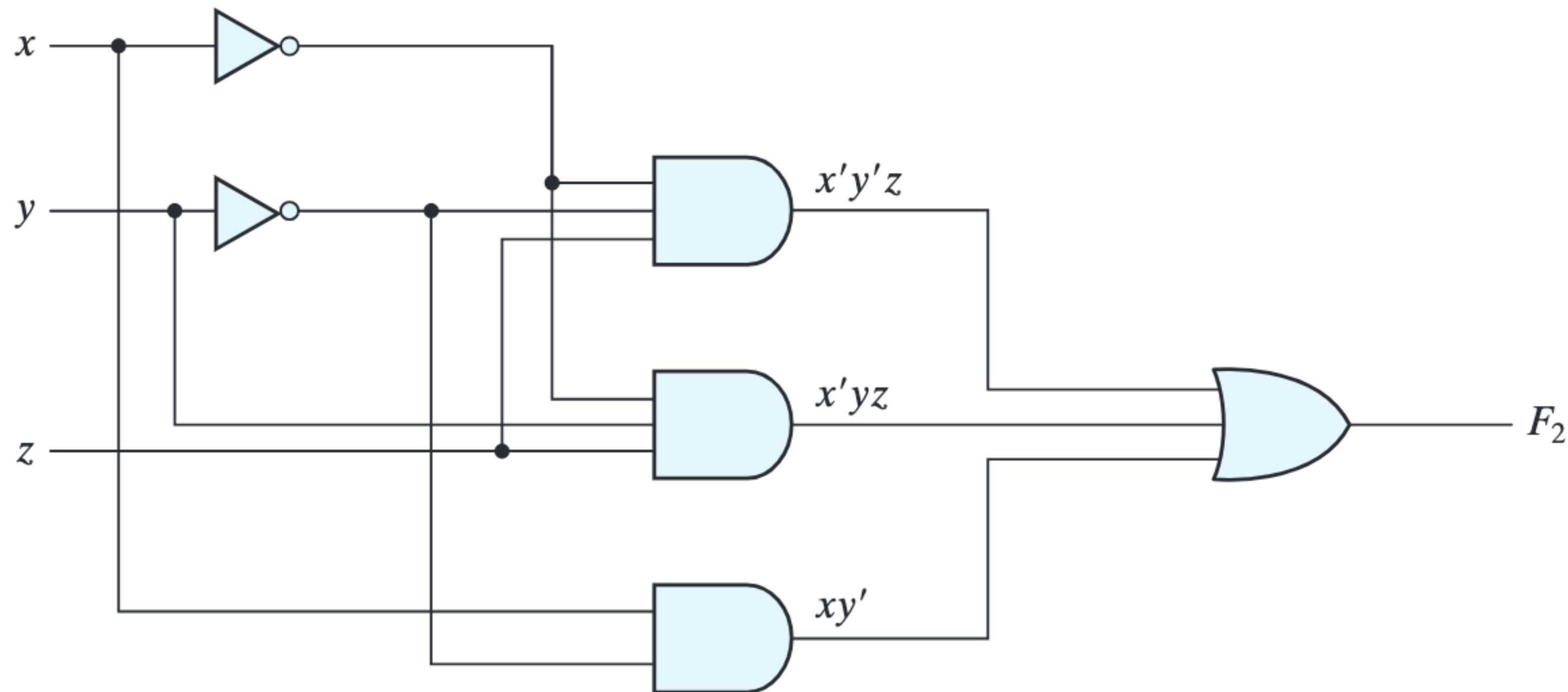
Complement of a function (2)

- $F' = (x'y + xy')'$
 - $(x'y)'(xy')'$
 - $(x+y')(x'+y)$
 - $xx' + xy + y'x' + y'y$
 - $xy + x'y'$



Truth tables and logic diagrams

- $F_2 = x'y'z + x'yz + xy'$



(a) $F_2 = x'y'z + x'yz + xy'$

x	y	z	F_2
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Simpler expressions lead to simpler circuits

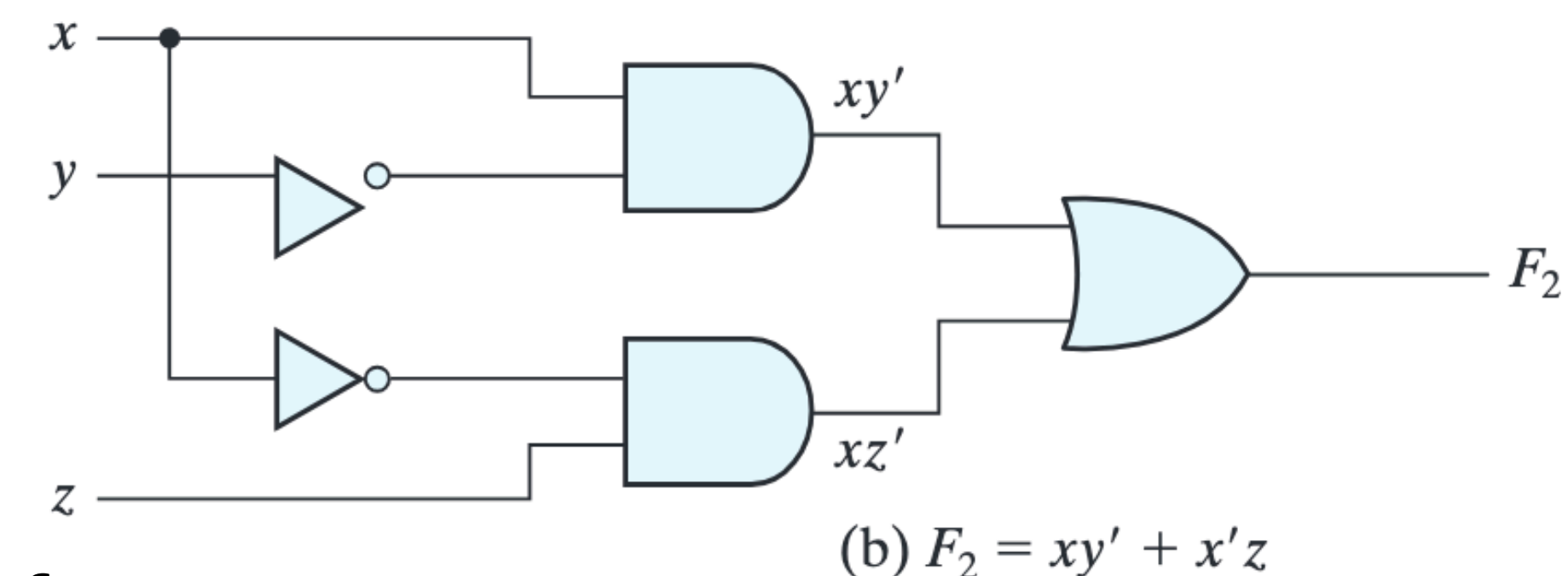
- There is only one way in which a function can be expressed in the truth table
 - But in a variety of ways in its algebraic and schematic form, all of which have equivalent logic

x	y	z	F_2
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

- $F_2 = x'y'z + x'yz + xy'$

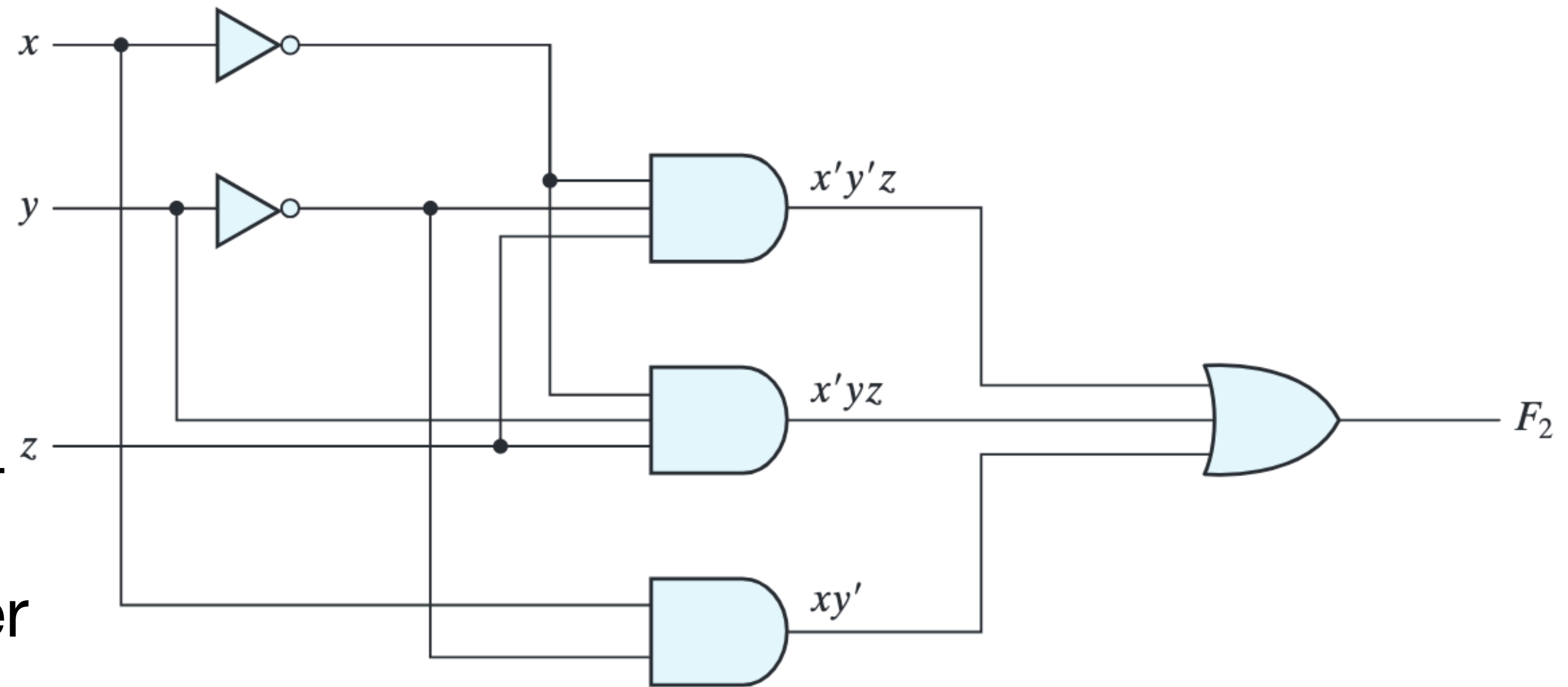
- $F_2 = x'z + xy'$

- Both produce the same truth table: same outputs for all possible input combinations. Both are equivalent

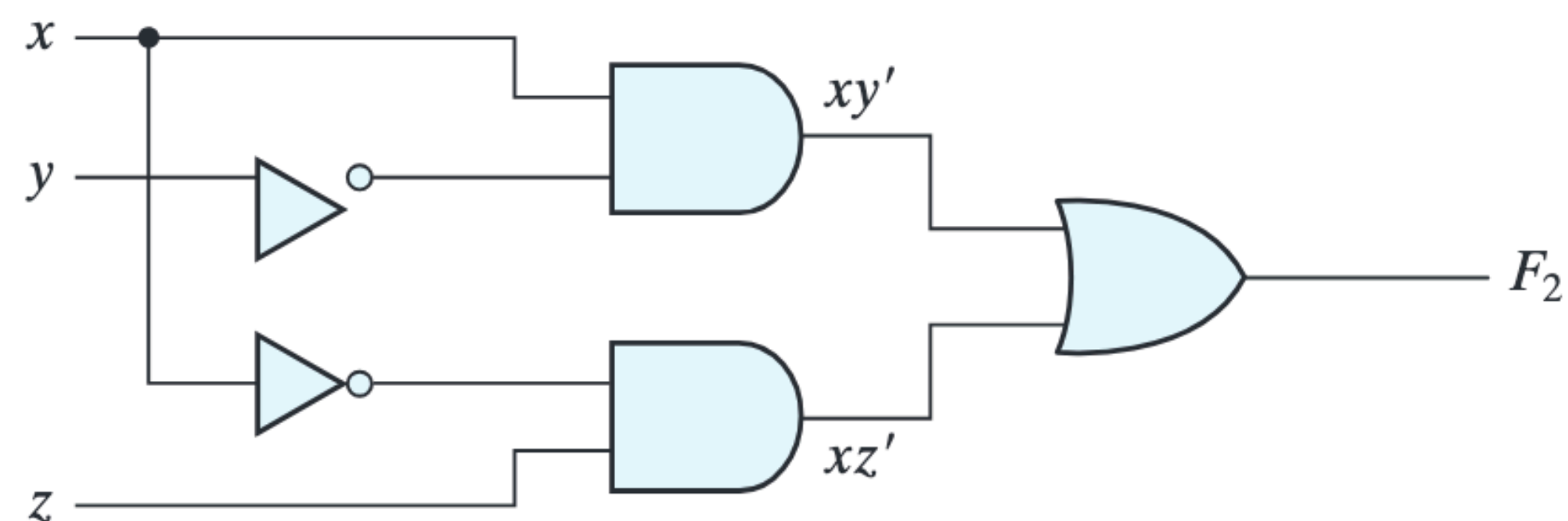


Algebraic Manipulations

- $F_2 = x'y'z + x'yz + xy'$. 3 terms and 8 literals.
- $F_2 = x'z + xy'$. 2 terms and 4 literals. Preferable because it requires fewer wires and fewer gates



(a) $F_2 = x'y'z + x'yz + xy'$



(b) $F_2 = xy' + x'z$

Simplify the following Boolean expressions

- $x(x' + y)$
 - xy
- $x + x'y$
 - $x + y$
- $(x + y)(x + y')$
 - x
- $xy + x'z + yz$
 - $xy + x'z + yz(x + x')$
 - $x(y + yz) + x'(z + yz)$
 - $xy + x'z$

Simplifying boolean functions

- Algebraically simplifying boolean functions is complicated
 - Approach: convert boolean function into its *canonical* form and then apply a “map method” to simplify
- $xy + x'z + yz$
 - $xy + x'z + yz(x + x')$
 - $x(y + yz) + x'(z + yz)$
 - $xy + x'z$

Minterms

- Call each row of the truth table as a *minterm*, represented by m_i where i represents the decimal equivalent of the binary number
- If you write the truth table of each minterm, notice that it outputs 1 in only its row of the truth table; other rows output 0

			Minterms	
x	y	z	Term	Designation
0	0	0	$x'y'z'$	m_0
0	0	1	$x'y'z$	m_1
0	1	0	$x'yz'$	m_2
0	1	1	$x'yz$	m_3
1	0	0	$xy'z'$	m_4
1	0	1	$xy'z$	m_5
1	1	0	xyz'	m_6
1	1	1	xyz	m_7

Canonical forms

Sum of minterms

- A boolean function can be expressed from a given truth table by forming a minterm for each combination of the variables that produces a 1 in the function and then taking the OR of all those terms.

<i>x</i>	<i>y</i>	<i>z</i>	Function <i>f</i> ₁	Function <i>f</i> ₂
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

- $f_1 = x'y'z + xy'z' + xyz = m_1 + m_4 + m_7$
- $f_2 = x'yz + xy'z + xyz' + xyz = m_3 + m_5 + m_6 + m_7$

Convert a function to sum of minterms from its expression

- $F = A + B'C$
- $A(B + B')(C + C') + (A + A')B'C$
- $ABC + ABC' + AB'C + AB'C' + AB'C + A'B'C$
- $ABC + ABC' + AB'C + AB'C' + A'B'C$
- $A'B'C + AB'C' + AB'C + ABC' + ABC$
- $m_1 + m_4 + m_5 + m_6 + m_7$
- $F(A, B, C) = \Sigma(1,4,5,6,7)$

Table 2.5
Truth Table for $F = A + B'C$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Maxterms

- Call each row of the truth table as a *maxterm*, represented by M_i where i represents the decimal equivalent of the binary number
- If you write the truth table of each maxterm, notice that it outputs **0** in only its row of the truth table; other rows output **1**

			Maxterms	
x	y	z	Term	Designation
0	0	0	$x + y + z$	M_0
0	0	1	$x + y + z'$	M_1
0	1	0	$x + y' + z$	M_2
0	1	1	$x + y' + z'$	M_3
1	0	0	$x' + y + z$	M_4
1	0	1	$x' + y + z'$	M_5
1	1	0	$x' + y' + z$	M_6
1	1	1	$x' + y' + z'$	M_7

Table 2.4
Functions of Three Variables

<i>x</i>	<i>y</i>	<i>z</i>	Function <i>f</i> ₁	Function <i>f</i> ₂
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Canonical forms

Product of maxterms

- A boolean function can be expressed from a given truth table by forming a **maxterm** for each combination of the variables that produces a **0** in the function and then taking the **AND** of all those terms.
- $f_1 = (x + y + z)(x + y' + z)(x + y' + z)(x' + y + z')(x' + y' + z) = M_0M_2M_3M_5M_6$
- $f_2 = (x + y + z)(x + y + z')(x + y' + z)(x' + y + z) = M_0M_1M_2M_4$

Minterms and Maxterms are complements

- $m'_j = M_j$

- $M'_j = m_j$

			Minterms		Maxterms	
x	y	z	Term	Designation	Term	Designation
0	0	0	$x'y'z'$	m_0	$x + y + z$	M_0
0	0	1	$x'y'z$	m_1	$x + y + z'$	M_1
0	1	0	$x'yz'$	m_2	$x + y' + z$	M_2
0	1	1	$x'yz$	m_3	$x + y' + z'$	M_3
1	0	0	$xy'z'$	m_4	$x' + y + z$	M_4
1	0	1	$xy'z$	m_5	$x' + y + z'$	M_5
1	1	0	xyz'	m_6	$x' + y' + z$	M_6
1	1	1	xyz	m_7	$x' + y' + z'$	M_7

Conversion between canonical forms

- $F(A, B, C) = \Sigma(1,4,5,6,7)$
- $F(A, B, C)' = \Sigma(0,2,3) = m_0 + m_2 + m_3$
- $F(A, B, C) = m'_0 m'_2 m'_3$
- $F(A, B, C) = M_0 M_2 M_3$
- $F(A, B, C) = \Pi(0,2,3)$

Interchange Σ and Π and list the numbers missing from the original form

Conversion between canonical forms

Why does the approach work?

- $F = xy + x'z$
- Find all ones from the truth table
 - $F = \Sigma(1,3,6,7)$
- Remaining have zeros in the truth table
 - $F = \Pi(0,2,4,5)$

Truth Table for $F = xy + x'z$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Minterms

Maxterms

Exercise: Identify midterms and maxterms

- $F = \Sigma(1,3,4,6) = \Pi(0,2,5,7)$
- Any Boolean function can be *uniquely* written in its canonical forms: sum-of-minterms or product-of-maxterms form

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Standard forms

Sum of products

- “Map method” will convert from canonical form to standard form, i.e., sum of minterms to sum of products
 - Literals: x , x' , y , y' , z , z' etc
 - Product terms: y' , xy , $x'yz'$
 - Sum of product: $y' + xy + x'yz'$

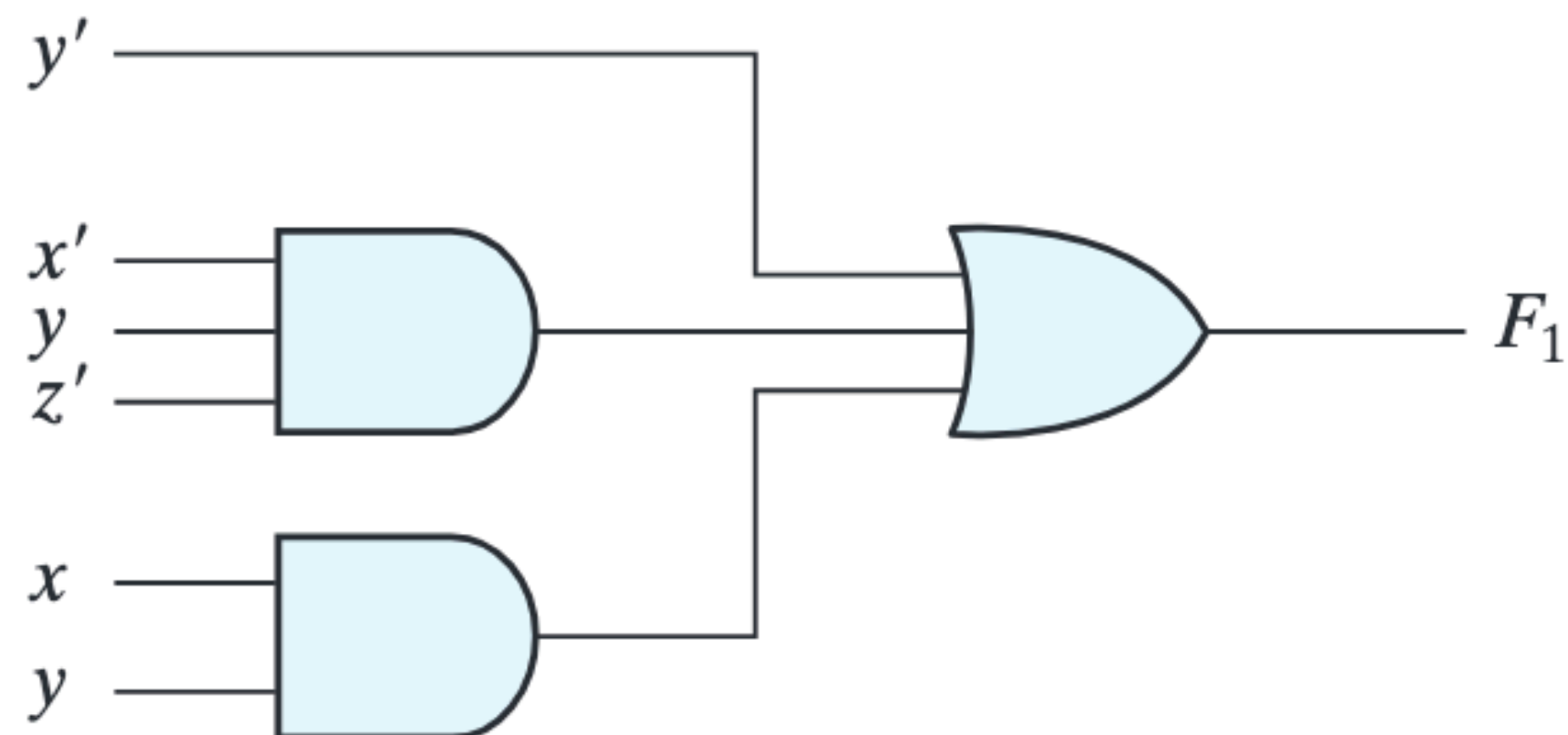
Standard forms

Product of sums

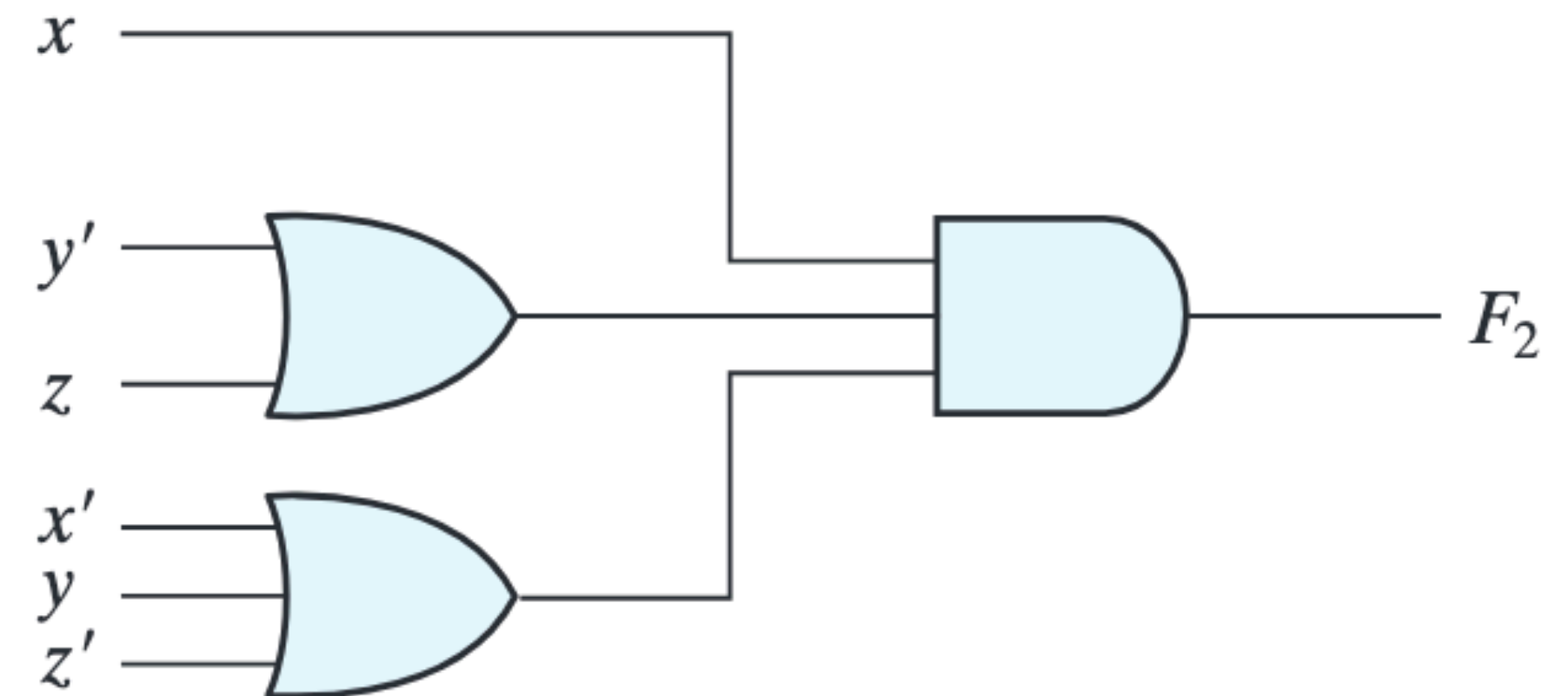
- “Map method” will convert from canonical form to standard form, i.e., product of maxterms to product of sums
 - Literals: x , x' , y , y' , z , z' etc
 - Product terms: x , $y'+z$, $x' + y + z'$
 - Sum of product: $x (y'+z) (x'+y+z')$

Standard forms lead to two-level structure of gates

- Preferred because it produces least amount of delay through the gates when the signal propagates from the input to output



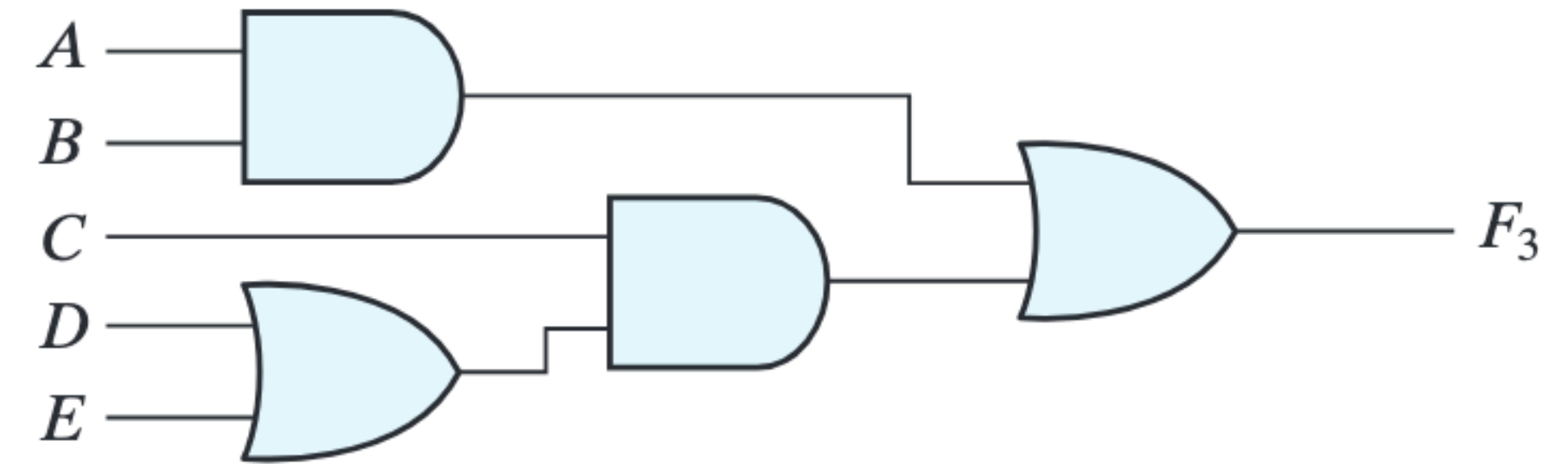
(a) Sum of Products



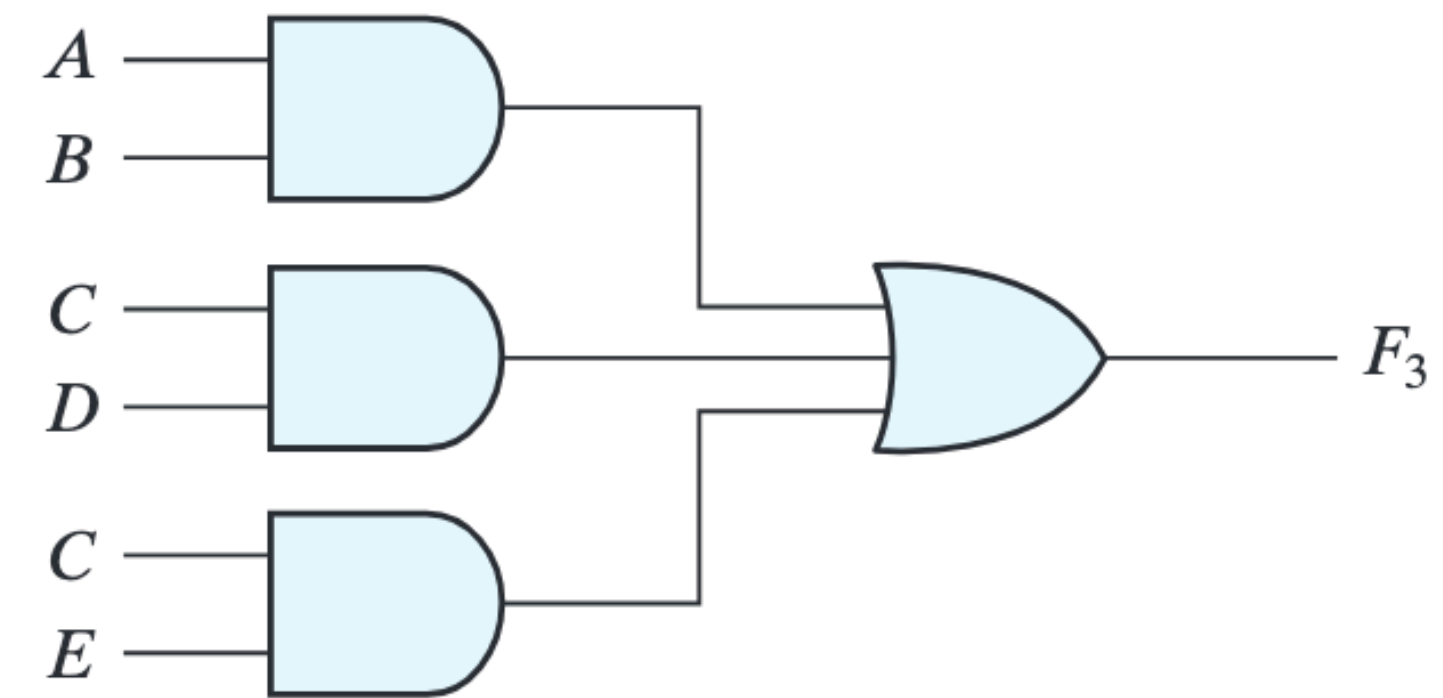
(b) Product of Sums

Non-standard forms lead to larger delays

- $F = AB + C(D + E)$
- $F = AB + CD + CE$



(a) $AB + C(D + E)$



(b) $AB + CD + CE$

Gate-level minimisation

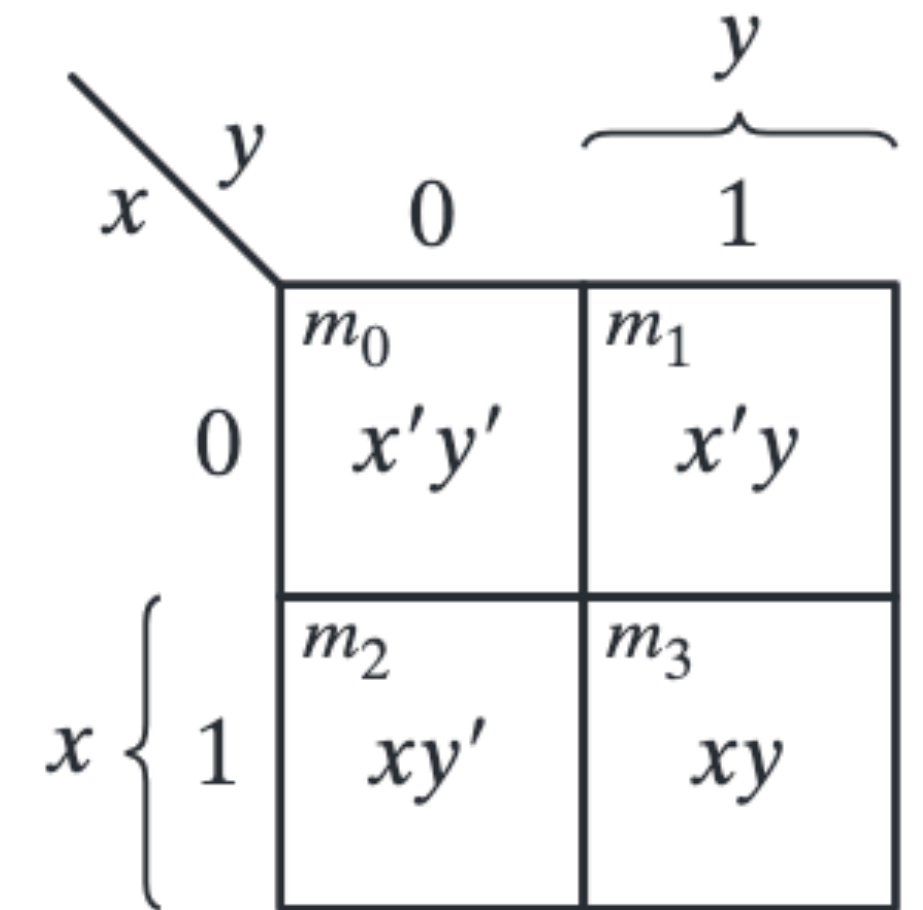
- Find an optimal gate-level implementation of the Boolean function
- Expressions can be simplified algebraically
 - For a given function, truth table is unique, but when expressed algebraically it can appear in many equivalent forms
 - $xy + x'z + yz$
 - $xy + x'z + yz(x + x')$
 - $x(y + yz) + x'(z + yz)$
 - $xy + x'z$
 - Lacks rules to predict each succeeding step
 - Unclear if we reached a “minimal function”
 - Least number of inputs/gates
 - Minimum number of terms with minimum number of literals in each term

Karnaugh map method

- Any Boolean function can be uniquely represented as sum of minterms
- Represent each Boolean function graphically
- Recognise patterns to derive alternative algebraic expressions for the same function. Select the simplest one
- Just follow the straightforward rules to recognise patterns
- Guaranteed to produce a minimal expression: minimum number of terms with minimum number of literals in each term

Two-variable K-map

0	0	m_0
0	1	m_1
1	0	m_2
1	1	m_3



- Four squares, one for each minterm
- 0 and 1 marked in each row and column designate the values of the variables.
- $x=0$ in first row. $x=1$ in second row. $y=0$ in first column. $y=1$ in second column

Representing functions in the K-map

- Basically a pictorial representation of the truth table
- xy
 - Equal to minterm 3 (= 11)
- Mark the minterms with 1 where the function outputs 1

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

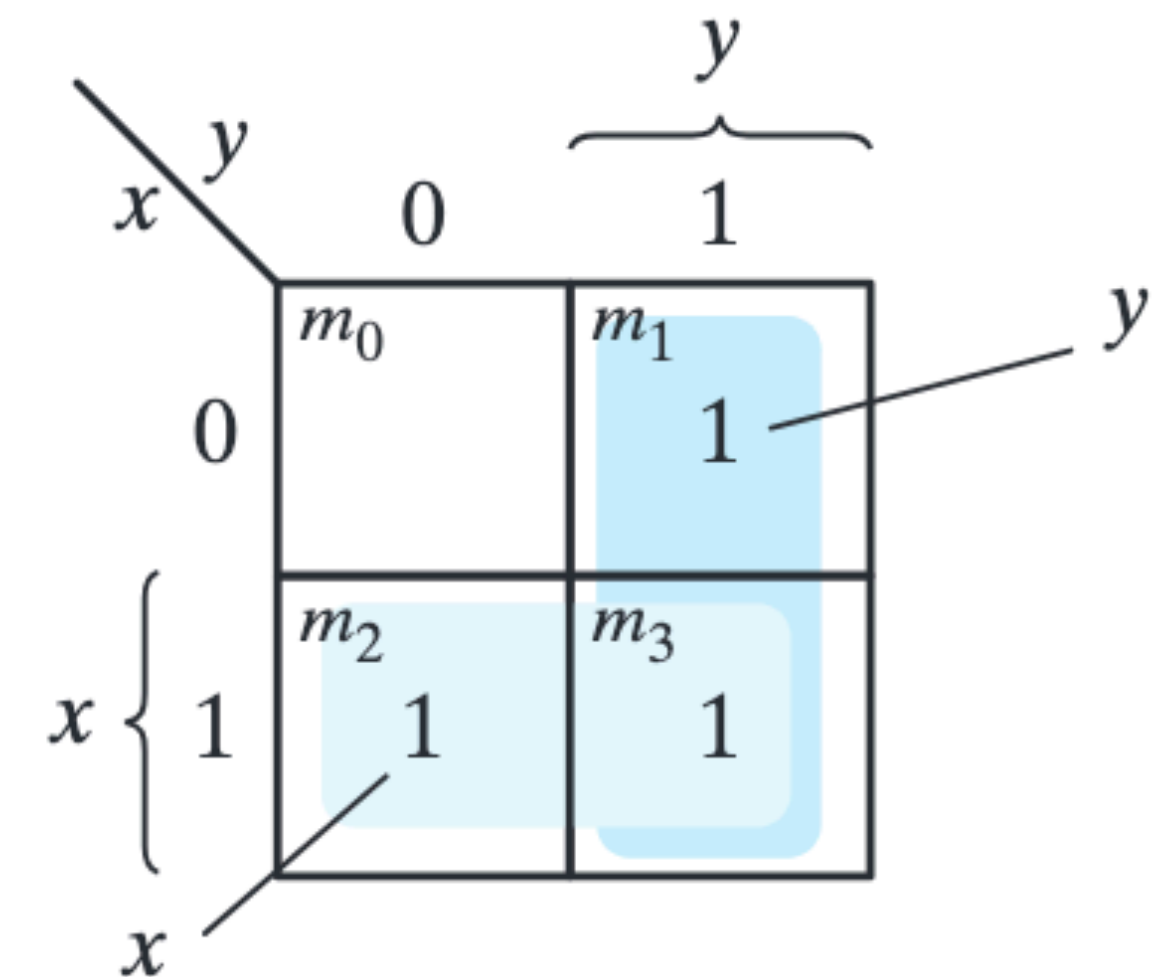
		y	
		0	1
x	0	m_0	m_1
	1	m_2	m_3 1

(a) xy

Representing functions in the K-map

- $x+y$
 - $m_1 + m_2 + m_3$
- The three squares could also be determined as the union of the squares of variable x in the second row and those of variable y in the second column

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1



(b) $x + y$

Three-variable K-map

$F(x, y, z)$

- m_5 (=101) is row 1 and column 01, i.e., $xy'z$
- Four squares for which each variable equals 1

		y			
		yz		11	10
x	0	m_0 $x'y'z'$	m_1 $x'y'z$	m_3 $x'yz$	m_2 $x'yz'$
	1	m_4 $xy'z'$	m_5 $xy'z$	m_7 xyz	m_6 xyz'

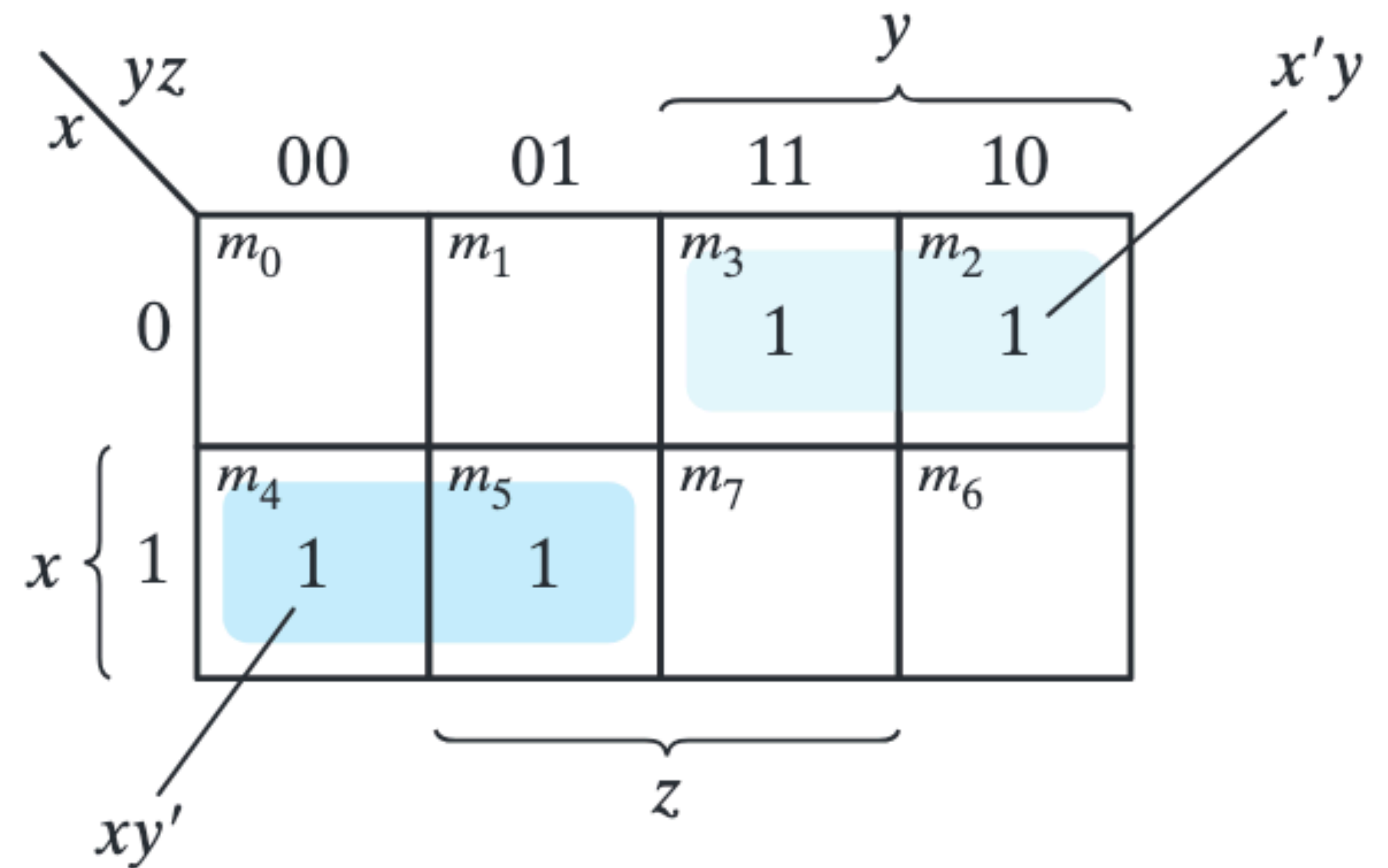
Usefulness of the map

- Only 1-bit changes between any two adjacent locations i.e., only one variable is primed in one square and unprimed in another, other variables are identical.
 - y' in m_5 and y in m_7 . Both have xz .
- Sum of two adjacent minterms will be one product term
 - $m_5 + m_7 = xy'z + xyz = xz(y + y') = xz$
- Any two minterms in adjacent squares (vertically or horizontally) when OR'ed together will remove the dissimilar variable

		y				
		yz		00	01	11
x	0	m_0 $x'y'z'$	m_1 $x'y'z$	m_3 $x'yz$	m_2 $x'yz'$	
	1	m_4 $xy'z'$	m_5 $xy'z$	m_7 xyz	m_6 xyz'	
		z				

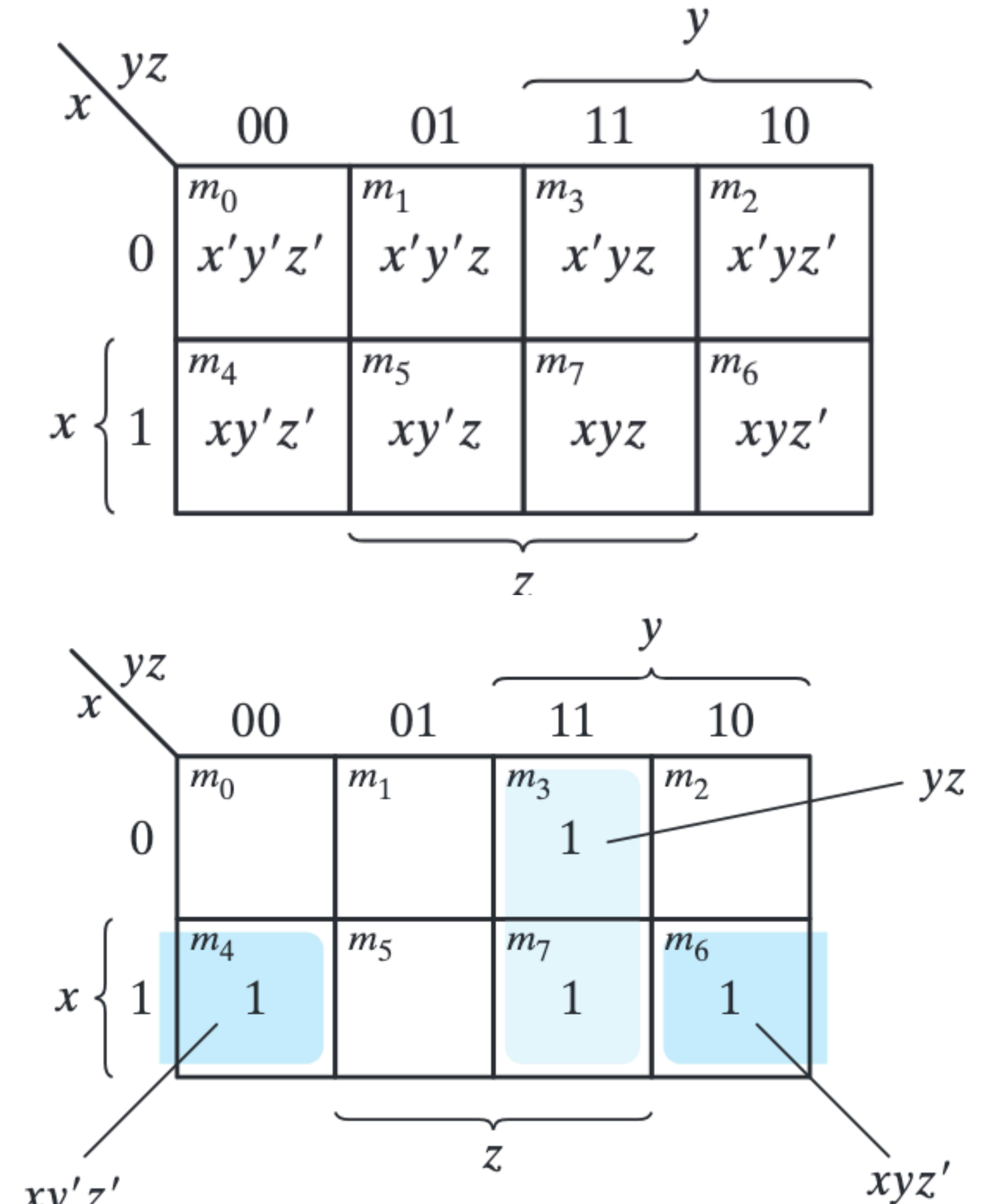
Simplifying functions

- $F(x, y, z) = \Sigma(2,3,4,5)$
- $F(x, y, z) = x'y + xy'$



Adjacent squares

- m_0 is adjacent to m_2 and m_4 is adjacent to m_6 because their minterms differ by only one variable.
- Simplify $\Sigma(3,4,6,7)$
 - $yz + xz'$



Note: $xy'z' + xyz' = xz'$

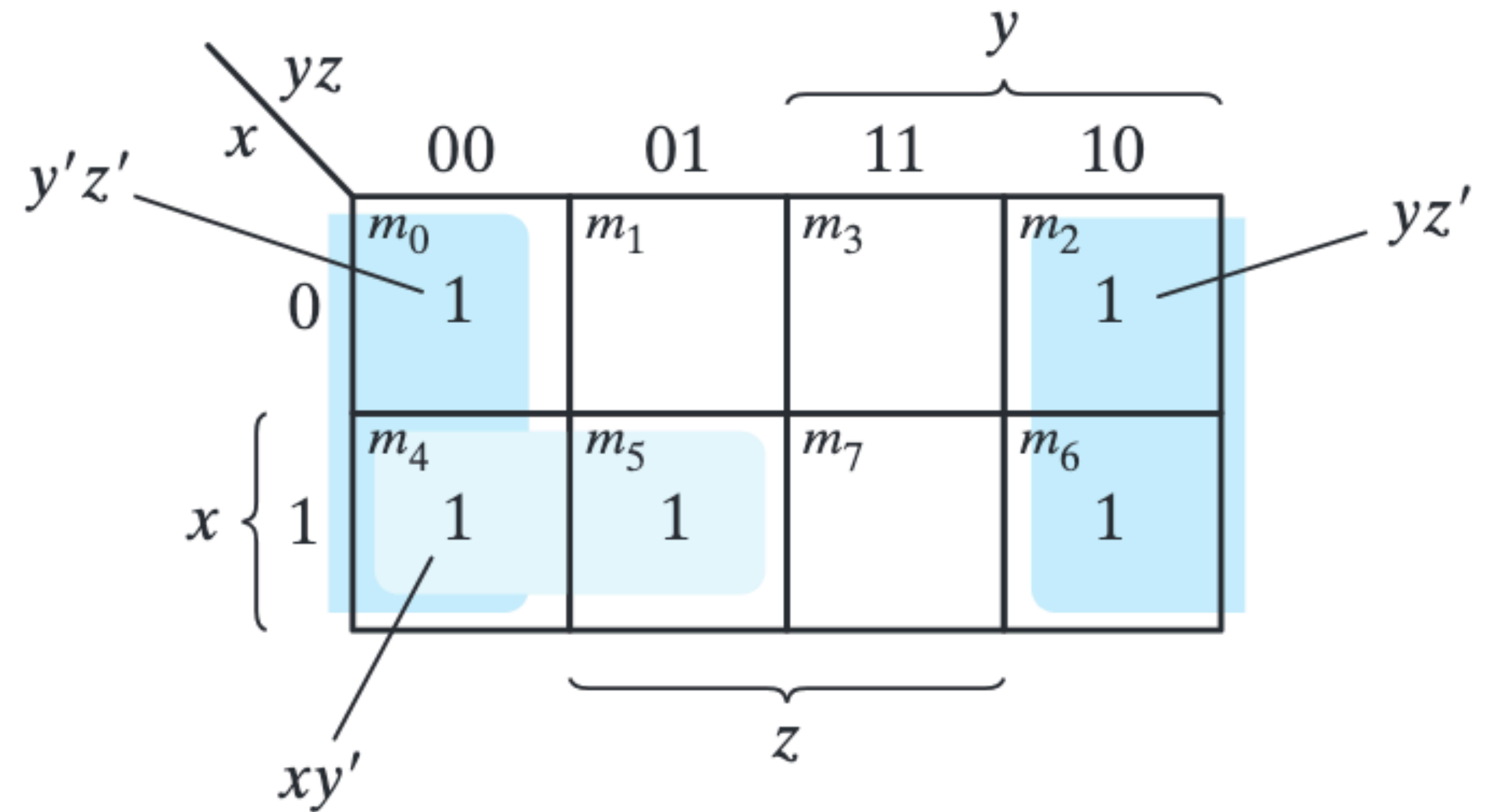
Four adjacent squares

- Any combination of four minterms results in an expression with only one literal
- $m_0 + m_2 + m_4 + m_6 =$
 - $x'y'z' + x'yz' + xy'z' + xyz'$
 - $x'z'(y' + y) + xz'(y' + y)$
 - $x'z' + xz'$
 - z'

		y			
		yz		11	10
x	0	m_0 $x'y'z'$	m_1 $x'y'z$	m_3 $x'yz$	m_2 $x'yz'$
	1	m_4 $xy'z'$	m_5 $xy'z$	m_7 xyz	m_6 xyz'

Simplify the Boolean function

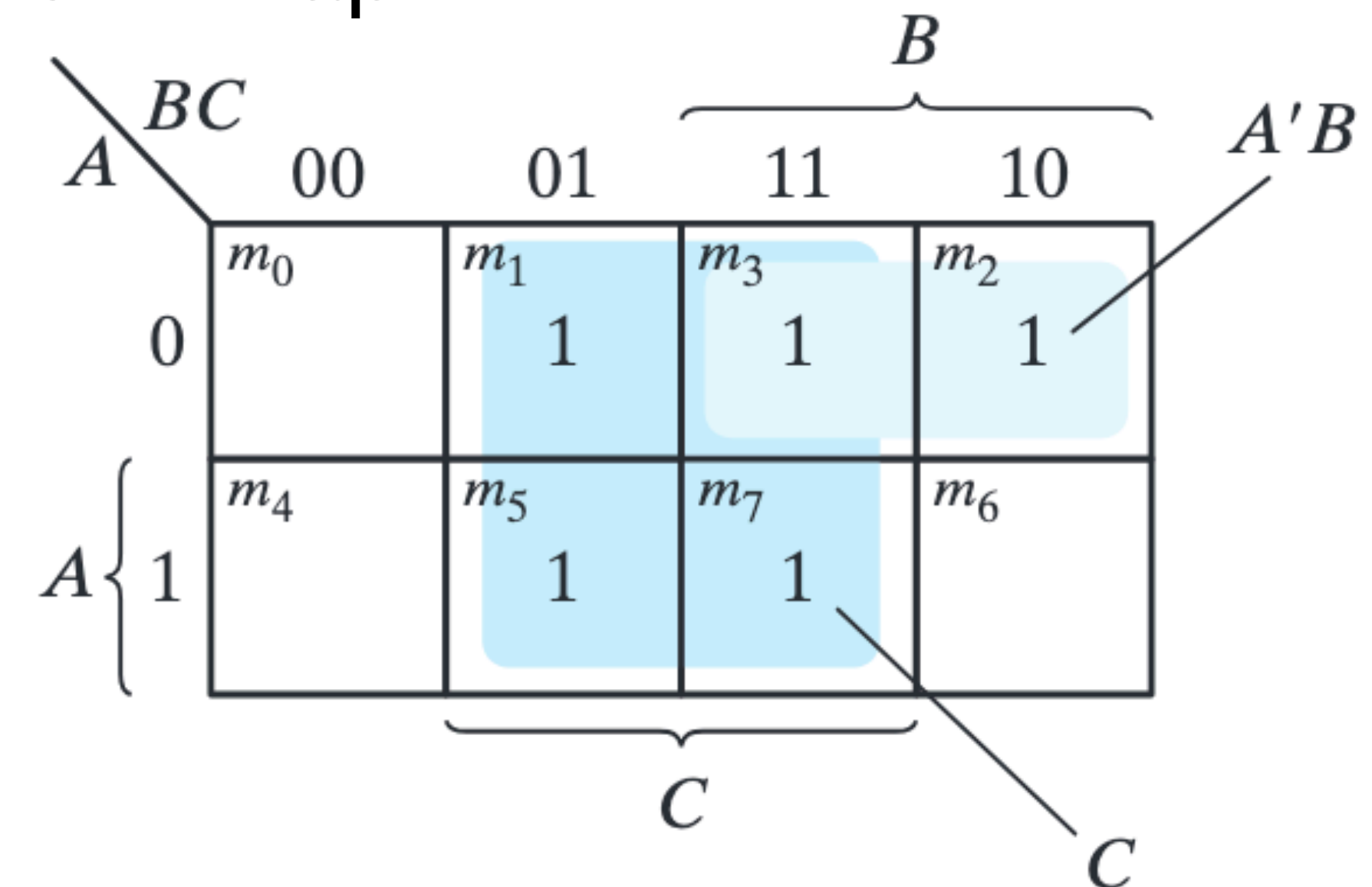
- $F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$
- $z' + xy'$
- Notice that m_5 is combined with an already used square m_4
- This is desirable as it represents xy' whereas the single m_5 square represents $xy'z$



Note: $y'z' + yz' = z'$

Simplify sum of products

- Function not in sum-of-minterms form: $F = A'C + A'B + AB'C + BC$
- Put 1 in all the squares corresponding to $A'C$; $A'B$; $AB'C$; BC
- Function has 5 minterms as it has 5 1's in the K-map
- $F = C + A'B$



Minimisation Approach

Combine adjacent squares in powers of two

- One square represents one minterm, giving a term with three literals
- Two adjacent squares represent a term with two literals
- Four adjacent squares represent a term with one literal
- Eight adjacent squares
 - Encompass the entire three-variable map and produces a function that is always equal to 1

Minimisation Approach (2)

- Cover all the minterms when we combine the squares
- Minimize the number of terms in the expression (try to combine more squares)
- There are no redundant terms (i.e., minterms already covered by other terms)
- Exercise: $F(x, y, z) = \Sigma(0, 2, 3, 4, 6)$
 - $z' + x'y$